In class lab #1 for Numerical Analysis

Your name: _____

scan and submit before 17:00

1. (5 points) What is the value of

```
>>  2.^(0:4)
```

going to be?

2. (10 points) Write the Matlab command that defines the function

$$u(x) = x^4 + 10x - 3 + \sin(x) + e^x$$

in a way that allows you to then write

```
>>  u([0 1 2 3 4])
```

Write the result of this simultaneous evaluation.

3. (10 points) Here are two functions in Matlab

```
>> f = @(x) (x.*(x+1))./((x+0.5).*(x+1.5));
>> g = @(x) (x./(x+0.5)).*((x+1)./(x+1.5));
```

(1) Write down the mathematical expression of these two functions.

(2) Give an explanation of why these two results differ:

```
>> f(10^200)
ans =
    NaN
>> g(10^200)
ans =
    1
```

`NaN` is Matlab for not-a-number. It is the typical Matlab answer to what you called an inderterminate result (or limit) in Calculus class $(0/0, \infty/\infty, \infty - \infty, 1^\infty, \cdots)$

4. (15 points) Find both roots of the quadratic equation $x^2 + 9^{12}x = 3$ with four-digit accuracy.

Step 1. Type

```
>> format long     %16 significant digits
```

The roots of a quadratic equation of form $ax^2 + bx + c = 0$ are given by the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{1}$$

Step 2. Please write the result in the command window.

```
>> x1 = ( -9^(12) - sqrt( 9^(24)+4*(3) ) )/2

x1 =


>> x2 = ( -9^(12) + sqrt( 9^(24)+4*(3) ) )/2

x2 =
```

If $4|ac| \ll b^2$, then b and $\sqrt{b^2 - 4ac}$ are nearly equal in magnitude, and **one of the roots is subject to loss of significance**.

We can find that $b = 9^{12}$ is positive. We recall that if b is **positive** in this situation, then the two roots should be calculated as

$$x_1 = -\frac{b + \sqrt{b^2 - 4ac}}{2a},\ x_2 = -\frac{2c}{(b + \sqrt{b^2 - 4ac})}. \tag{2}$$

Step 3. Please write the result in the command window.

```
>> x2 = ( -2*(-3) )/( 9^(12) + sqrt( 9^(24)+4*(3) ) )

x2 =
```

5. (15 points) Similar to the above case, if b is **negative** and $4|ac| \ll b^2$, please find two roots of $ax^2+bx+c = 0$ to avoids the loss of significance problem.

6. (20 points) Calculate the expressions that follow in double precision arithmetic (using Matlab, for example) for $x = 10^{-1}, 10^{-2}, \cdots, 10^{-10}$. Then, write down an alternative form of the expression that doesn't suffer from subtracting nearly equal numbers and use it to repeat the calculation and make a table of results. Report the number of correct digits in the original expression for each x.

$$(a)\ \frac{1-\sec x}{\tan^2 x} \qquad (b)\ \frac{1-(1-x)^3}{x} \qquad (c)\ \sqrt{x^2+1}-1 \qquad (d)\ \sin(x)-x$$

7. (10 points) Find an alternate forms of the following formulas when x is a very large number.

$$(a)\quad \sqrt{x^2+1}-x \qquad (b)\quad \ln(x+1)-\ln(x)$$

8. (15 points) Evaluate the number of the arithmetic operations of $P(x)=a_nx^n+a_{n-1}x^{n-1}+\cdots+a_1x+a_0$. Design an algorithm to reduce the number of this operations of this calculation. And solve the following the polynamial $P(x)=23x^9+47x^7+68x^6+107x^3+98$ with $x=4$ by your algorithm.